

Jack Goh Pin Kai

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Professional Summary

Computer Science undergraduate with experience in software development, web development, data analysis, database systems, and application development. Proficient in C++, C#, Python, SQL, and Java, with project experience in full-stack development, RESTful APIs, relational databases, data modelling, and real-time graphics programming.

Key Skills

Programming: C++, C#, Java, JavaScript, Python, PHP, SQL

Data Analytics: Data Analysis, Relational Databases, Database Design, Data Modelling

Software Development: React.js, HTML, JSON, RESTful APIs, Git, Visual Studio

Graphics/Simulation: OpenGL, GLSL, ImGui, ECS

Tools: Microsoft Office, Adobe Creative Suite, PowerApps

Projects

NYP Mobile Events Manager Application

2019 – 2019

Software Developer

- Analysed user requirements and designed an event management application for users to view, host and register for events.
- Designed a MySQL relational database and developed RESTful APIs using Laravel and PHP, including authentication and database integration.
- Built a React.js frontend and integrated it with backend APIs for data retrieval and event management.

[Pulse Protocol](#)

2025 – 2026

Graphics Programmer

- Developed core 2D graphics systems for a custom C++/OpenGL engine, integrating ECS components with the rendering pipeline.

- Implemented a batched sprite renderer with texture/shader grouping, layer-based draw ordering, sprite-sheet animation, alpha blending, and GLSL shaders to reduce unnecessary OpenGL state changes and draw calls.
- Built editor and gameplay camera systems supporting pan, zoom, coordinate transformations, viewport resizing, and 16:9 letterboxing.
- Implemented texture and asset loading, FreeType font rendering, UI/debug text, and AABB/debug visualization tools.
- Integrated graphics features into an ImGui-based editor and collaborated with programmers, designers, and artists to deliver a polished playable game.

Work Experience

Sakura System Services

2022 - 2023

Systems Engineer

- Developed and maintained web-based systems based on client business and technical requirements.
- Diagnosed application issues and implemented solutions to support production systems.
- Supported deployment, maintenance and troubleshooting of customer web applications.

Singapore Armed Forces

2020 – 2022

Military Specialist

- Led and supported personnel while coordinating operational responsibilities in time-sensitive environments.
- Applied structured decision-making, teamwork and communication during operational and emergency situations.

Sopra Steria

2019 – 2019

Android Developer Intern

- Developed and supported android applications using Android Studio based on project and client requirements.
- Collaborated with team members to identify and resolve project workflow and application development issues.
- Prepared technical and project presentations for internal stakeholders and client-facing discussions.
- Communicated with clients and team members to clarify requirements, provide updates, and support project delivery.

Education Summary

Singapore Institute of Technology & DigiPen Institute of Technology

Bachelor of Science with Honours in Computer Science in Real-Time Interactive Simulation
Expected Graduation: 2028

Nanyang Polytechnic
Diploma in Multimedia & Infocomm Technology
Graduated: 2020

Licenses & Certifications

AI Singapore

Issued Jun 2025

AI4I – Literacy in AI

Credential ID 83957AC035-7F119FDAFC-156A3145B

Co-Curricular Activities & Awards

DigiPen Institute of Technology

Overseas Immersion Programme (OIP)

- Studied in Europe-Bilbao campus in Spain for one trimester
- Interacted with European professors and mentors while experiencing life in a different culture with overseas peers

Nanyang Polytechnic (NYP)

Chinese Cultural Club (Committee)

- Supported club and cultural activities through booth operations, visitor engagement, event coordination, and cultural showcases, including MUSE and calligraphy events.
- Collaborated with committee members to support event preparation and participant experience.

FLCD Event

- Supported programme execution as **cue personnel**, coordinating event timing, transitions, and backstage activities.
- Assisted with booth operations and event logistics while collaborating with team members to ensure smooth event delivery.

Hamamatsu 2019 Exchange Event

- Supported Japanese exchange students in adapting to the local environment and participating in programme activities.
- Applied cross-cultural communication and interpersonal skills to build rapport and create a welcoming experience.

Edusave Good Progress Award

- Issued by Ministry of Education (MOE)
- Students who are within the top 10% of their institution's level and course in terms of improvement in academic performance and have demonstrated good conduct.